of probabilities (linear operators instead of measures), and it takes its illustrating applications seriously enough to explain some of the underlying physical theory. This makes it an excellent source book for teachers. On the other hand, if the book were to be considered as a text at a North American university, I am not sure at which level it would fit. The text moves a bit fast for undergraduates, although it could be an exciting possibility for a strong class of mathematics or engineering seniors. Perhaps it is best suited for first-year graduate students in statistics or applied mathematics, needing a reasonable background in probability with stochastic processes but perhaps not a full-blown measure theoretic course.

After Chapter 1 on "Uncertainties, Intuition, and Expectation," the mathematical treatment starts in Chapter 2. An expectation is a rule for determining a number $\mathbf{E}(X)$ from a numeric-valued function $X(\omega)$, called an *observable,* or random variable. One starts with expectation values for a family $\mathcal{Y}$ of random variables Y. From this information, the purpose is to determine as accurately as possible the expected value of other random variables in accordance with the axioms defining an expectation: a linear operator, normalized so that $\mathbf{E}1 = 1$ and continuous in the sense that if $X_n(\omega)$ increase monotonically to a limit $X(\omega)$, then $\mathbf{E}(X) = \lim_n \mathbf{E}(X_n)$. Among the immediate applications of this concept are expected profit optimization for a news agent, sample survey formulas for the mean and variance of a sample mean, and least squares estimation of random variables.

Chapter 3 defines probability as the expected value of an indicator function. The Kolmogorov axiomatization is an immediate consequence of the axiomatization of the previous chapter. Although expectations were the natural approach to probability in the early days, it appears that the St. Petersburg paradox made the expectation approach fall out of fashion. Kolmogorov's axiomatization, using measure theory, was in answer to a question by Hilbert on the 1900 Mathematics Congress in Paris, where he said that "as for the axioms of the probability calculus, it seems desirable that any logical study of them should go hand in hand with a rigorous and satisfactory development of the method of mean values in mathematical physics." In quantum probability theory, von Neumann produced such an axiomatization the year before Kolmogorov's *Grundbegriffe der Wahrscheinlichkeitsrechnung!*

Chapter 4 introduces some of the standard models. The approach is different from the usual one. For example, the multinomial distribution does not come out of a combinatorics argument. Rather, using an approach common in statistical physics, the author starts with a microdescription of distributing N molecules over M cells, with the M^N possible configurations being equally probable. Let the random variable ζ_k take the value w_j if the kth molecule falls in the jth cell. Then

$$\mathbf{E}(\zeta_k) = \frac{1}{M}\sum_{j=1}^{M} w_j, \qquad k = 1, \ldots, N.$$

If N_j of the ζ_k take value w_j, then we must have

$$\prod_{k=1}^{N} \zeta_k = \prod_{j=1}^{M} w_j^{N_j}.$$

The random variables N_j constitute a macro-description of the system, where individual molecules no longer are identifiable. Taking expectations, we see that

$$\mathbf{E}\left(\prod_{j=1}^{M} w_j^{N_j}\right) = \mathbf{E}\left(\prod_{k=1}^{N} \zeta_k\right) = \prod_{k=1}^{N} \mathbf{E}(\zeta_k) = \left(\sum_{j=1}^{N} w_j/M\right)^{N}.$$

Letting the possible values of $\mathbf{N} = (N_1, \ldots, N_M)$ be denoted $\mathbf{n} = (n_1, \ldots, n_M)$, we can identify $\mathbf{E}\prod_j w_j^{N_j}$ with $\sum_{\mathbf{n}} \mathbf{P}(\mathbf{n}) \prod_j w_j^{N_j}$ to get the multinomial probability formula for $\mathbf{P}(\mathbf{n})$.

Another approach, often used in this book to obtain distributions, is through conditioning. An alternative derivation of the multinomial distribution is as a conditional distribution of independent Poisson variates, constrained to sum to N. The hypergeometric distribution is similarly obtained through conditioning from two independent binomial variates. Chapter 5 is devoted to conditioning, introducing conditional expectations as random variables, with applications to statistical decision theory, communication theory, and acceptance sampling. Chapter 6 studies the special case of independence, yielding renewal theory, recurrence, branching processes, and a fairly rigorous derivation of a Gibbs distribution.

Limit theory in the expectation approach is naturally based on characteristic functions. Chapter 7 deals with the weak law of large numbers and the central limit theorem. Chapter 8 covers continuous distributions and transformations. An interesting question here is how to deal with the Cauchy distribution. Because it does not have an expectation, it would superficially not fall into the framework outlined earlier. The simplest way around this would be to look separately at positive and negative parts (which have infinite expectations, as is allowed), but the author seems to prefer deriving the Cauchy distribution as the x coordinate of the point of impact of a photon, emitted from the origin, on the line $y = a$. I would have liked the author to emphasize more clearly the problem of how random variables with infinite expectation fall into the theory.

Chapter 9 treats Markov processes in discrete time. Among this chapter's interesting features is the introduction of the generator, which, although perhaps not strictly necessary here, simplifies the introduction of continuous time Markov processes. The chapter presents a thorough discussion of laws of detailed balance, or reversible chains. The approach used to deal with ergodic results only manages to show Abel convergence of the n-step transition probabilities, whereas a difficult Tauberian argument would be needed to verify convergence of the transition probabilities themselves.

For an undergraduate course, I would stop here; the rest of the book is considerably harder. Chapter 10 studies Markov processes in continuous time, with an excellent discussion of reversibility and the need for this in physical systems. The chapter elucidates the importance of the Ehrenfest model in providing acceptance of Boltzmann's statistical physics.

After Chapter 11, on second-order theory, Chapter 12 deals with consistency and extension results in the finite dimensional case. Here convexity plays an important role. For example, the possible values of $\mathbf{E}(X)$ (varying the operators $\mathbf{E}$) is the closed convex hull of the possible values of $X(\omega)$. The chapter also derives bounds on the value of $\mathbf{E}(X)$ from knowing $\mathbf{E}(Y)$ for Y in a family $\mathcal{Y}$.

Chapter 13 covers the usual concepts of convergence and their relations. Chapter 14 discusses martingales, including a strong law for random variables with constant conditional mean $\mathbf{E}(Y_n \mid Y_1, \ldots, Y_n)$, and Chapter 15 deals with extensions to the infinite-dimensional case—in particular, weak convergence. The final chapter, 16, looks at particular applications to information theory, dynamic programming, and quantum probability. This last application vividly illustrates the usefulness of the expectation approach to probability.

The book contains many exercises and comments. Although these are somewhat short of the type of elementary calculations that students often need to grasp new concepts, there is sufficient interesting material to use as the basis for homework. Among the less common exercises are questions on the distinction between continuous and thermodynamic limits for the spatial distribution of gas molecules, the distribution of size of polymer molecules, reversibility for deterministic dynamic systems, the Kalman filter, principal components, and data compaction versus data compression.

I am certainly going to borrow freely from *Probability Via Expectation* in my teaching. I am less certain that I would use it as a sole text. Nevertheless, I recommend it wholeheartedly to all those probability teachers on the lookout for serious applications of the theory.

PETER GUTTORP
University of Washington

Statistical Reasoning With Imprecise Probabilities.

Peter Walley. New York: Chapman and Hall, 1991. xii + 706 pp. $137.

This book develops a general theory of upper and lower probabilities, and everyone interested in the foundations of statistical inference should own a copy despite the high price. The book is at once a theory of partial belief, a formalization of Bayesian robustness, and a thorough (if dogmatic) critique of standard Bayesian statistics. The author's thesis is that imprecision of probability is a concept that can be made precise and that the "dogma of precision" is an article of faith in the Bayesian paradigm that has led modern Bayesian statistics astray. The book is more than 700 pages long and was more than 10 years in the making. To do it justice requires providing some background on subjective probability.

The best and simplest justification for the laws of probability is the betting argument of Ramsey, Shimony, and De Finetti. Let X be a random variable and let $P(X)$ be the price at which an observer is indifferent between buying and selling X (regardless of other transactions). $P(\cdot)$ is called coherent if it is not possible to inflict a sure loss on the observer using bets at the stated prices. A necessary and sufficient condition for P to be coherent, assuming that the observer is obligated to accept combinations of bets, is that P be an expectation operator. The restriction of P to indicator functions is a probability, in the usual sense. (I am ignoring delicate issues about finite additivity.)

But what if an observer cannot specify P exactly? This uncertainty about P seems to be of a different nature than the uncertainty about X. A close look at the coherence argument shows that it is possible to separate these two types of uncertainty by using a set M of probabilities and defining a lower expectation $\underline{P}(X) = \inf_{P\in M} P(X)$ and an upper expectation $\bar{P}(X) = \sup_{P\in M} P(X)$. This second type of uncertainty is represented by the fact that $\bar{P}(X) - \underline{P}(X)$ may be positive. But what justification is there for replacing a precise expectation with an upper and lower expectation? Such a justification can be found by interpreting $\underline{P}(X)$ as the supremum of the prices at which the observer would buy X and $\bar{P}(X)$ as the infimum of the prices at which the observer would sell X. Then we say that $\underline{P}$ is coherent if (a) $\underline{P}(X) \geq \inf X$, (b) $\underline{P}(\lambda X) = \lambda \underline{P}(X)$ for $\lambda > 0$, and (c) $\underline{P}(X + Y) \geq \underline{P}X + \underline{P}Y$. Condition (c) is justified by the following argument. For any $\varepsilon > 0$, I can force you to pay $\underline{P}X - \varepsilon/2$ for X and $\underline{P}Y - \varepsilon/2$ for Y. Hence I can force you to pay $\underline{P}X + \underline{P}Y - \varepsilon$ for $X + Y$. Thus $\underline{P}(X + Y)$ must be greater than or equal to $\underline{P}X + \underline{P}Y$. A fundamental theorem is that $\underline{P}$ is coherent if and only if there exists a nonempty, weak* closed, convex set of probabilities M such that $\underline{P}X = \inf_{P \in M} PX$ and $\bar{P}X = \sup_{P \in M} PX$ for all X. Thus coherence implores us to use sets of probabilities. When restricted to indicator functions, these expectations are called upper and lower probabilities and we write $\underline{P}(A)$ and $\bar{P}(A)$ for $\underline{P}(I_A)$ and $\bar{P}(I_A)$, where I_A is the indicator function for A. In general, $\underline{P}(X) = 1 - \bar{P}(-X)$, so it is possible to consider just $\underline{P}$, say.

Researchers in robustness have used sets of probabilities for a long time. In frequentist theory, Huber (1981) and Huber and Strassen (1973), among others, developed the mathematical tools to do this. Bayesian theory based on sets of probabilities has a long history dating back to Keynes, Koopman, and Good. Now known as robust Bayesian inference, this has attracted much attention due to the work of Berger (1984, 1990) and DeRobertis and Hartigan (1981). *Statistical Reasoning With Imprecise Probabilities* is a formalization of robust Bayesian theory and for this reason alone is a major contribution. But in fact, Walley downplays this feature, and it is the way that he departs from standard robust Bayesian inference that makes the book provocative. To explain this, we need to distinguish between two interpretations of imprecise probabilities: the "sensitivity analysis interpretation" and the "direct" interpretation, as identified by Walley. Levi (1985) called these "imprecision" and "indeterminacy." Walley considers several other interpretations as well.

The "sensitivity analysis" interpretation of upper and lower probabilities is that probability theory is the correct expression of uncertainty and that upper and lower probabilities arise only because of our inability to exactly express the probabilities. To put it briefly, we act as if there is a true probability P somewhere in M. This is the interpretation embraced by most robust Bayesians because most Bayesians accept what Walley calls the "dogma of ideal precision," which asserts that "in any problem, there is an ideal probability model which is precise, but which may not be precisely known" (p. 7). Walley eschews this assumption. The "direct" interpretation is that upper and lower probabilities are the fundamental expressions of uncertainty—there is no true P in M. To the sensitivity analyst, the laws of probability are correct, but the probabilities assigned in a given application might not be. In the direct interpretation, even the usual laws of probability themselves are questionable. (This viewpoint was taken to its greatest extreme by Fine (1988), who abandoned even the set M of dominating probabilities.) Because of our backgrounds and training, most of us will quickly reject the direct interpretation and accept the dogma of ideal precision. But it is worth reflecting on the fact that the entire edifice of modern statistics, Bayesian and frequentist alike, is implicitly built on this assumption.

The differences between these two interpretations are not trivial. Consider, for example, a finite set Ω and suppose that our uncertainty is permutation invariant. The sensitivity analyst would use a set of permutation-invariant probabilities to model this. Of course, there is only one such probability: the uniform. The direct approach requires instead that $\underline{P}$ be permutation invariant. As it turns out, there is a rich collection of such lower probabilities besides the uniform (Wasserman and Kadane 1992). In fact, if we start with any collection of probabilities M and form the set M' of all permutations of the mass functions in M, we get a lower probability $\underline{P}$ that is permutation invariant. Clearly, these lower probabilities will be much different than the uniform probability. Thus the way we model structural constraints depends on which interpretation of imprecise probabilities we use. Another example of the difference between the two schools of thought occurs when we construct sets of priors in location problems. Taking the direct approach, Pericchi and Walley (1991) considered certain translation-invariant lower probabilities. The corresponding classes of priors turned out to be very different from the classes of priors used in the robust Bayesian literature. As Pericchi and Walley stated, they wanted "reasonable classes of priors," whereas the sensitivity analyst looks for "classes of reasonable priors."

Statistical Reasoning With Imprecise Probabilities contains some discussion about finitely additive and countably additive probabilities. In particular, Walley insists that expectations be conglomerable. This entails that if $B_1, B_2, \ldots$ is a denumerable partition such that $\underline{P}(B_n) > 0$ and $\underline{P}(B_n X) \geq 0$ for all $n \geq 1$, then $\underline{P}(X) \geq 0$. In words, if a gamble X is conditionally desirable given each element of the partition, then the gamble X is desirable unconditionally. When applied to precise probabilities, this means (ignoring some technical details) that the probability must be countably additive (Schervish, Seidenfeld, and Kadane 1984). Thus Walley declares that finitely additive probabilities are not rational, at least in the case of precise probabilities. Curiously, there are sets of finitely additive probabilities that generate lower probabilities that are conglomerable, once again underscoring the difference between the two interpretations of imprecise probabilities. (Unlike sensitivity analysts, those following the direct interpretation are untroubled by abberant behavior of the dominating measures.)

Walley's requirement of conglomerability seems reasonable to me. Otherwise, strange things happen. Consider a finitely additive (and hence nonconglomerable) probability that makes $P(A) = \frac{1}{4}$ for some event A but $P(A \mid B_i) = \frac{3}{4}$ for all B_i. If I know that I will set $P(A \mid B_i) = \frac{3}{4}$ no matter what B_i occurs, why not just set $P(A) = \frac{3}{4}$ now? Something is amiss. Some of my finitely additive friends disagree. Indeed, they are willing to accept the peculiarities of nonconglomerability and hence would reject Walley's conglomerability restriction.

But upper and lower probabilities have their own strange behavior. Specifically, they display "dilation." By this I mean that $\underline{P}(A \mid B) < \underline{P}(A) \leq \bar{P}(A) < \bar{P}(A \mid B)$ for every B in the partition $\mathcal{B}$. In extreme cases we might have $\underline{P}(A) = \bar{P}(A) = \frac{1}{2}$, say, but $\underline{P}(A \mid B) = 0$ and $\bar{P}(A \mid B) = 1$ for all $B \in \mathcal{B}$. This is unsettling for the same reason that nonconglomerability is disturbing. It seems that we are better off not learning B at all. Walley (pp. 298–299) does not find this bothersome, and I suppose it is a matter of taste. Seidenfeld and Wasserman (1993) showed that dilation is a common phenomenon for upper and lower probabilities, so it cannot be written off as a pathology. Dilation brings the conflict between the two interpretations of upper and lower probabilities to the fore. The direct approach is stuck with dilation, while the sensitivity analyst can always claim that underneath it all is a perfectly respectable, well-behaved probability.

A nice feature about upper and lower probabilities is that there exists a transformation-invariant expression of ignorance, a holy grail in Bayesian statistics. To represent ignorance, we use the set $\mathcal{P}$ of all probabilities. This gives the vacuous upper and lower probability for which $\underline{P}(A) = 0$ if $A \neq \Omega$ and $\underline{P}(A) = 1$ if $A \neq \varnothing$. It is tempting to conclude that this solves the problem of finding an objective prior for Bayesian inference. But a vacuous prior gives a vacuous posterior, no matter how much data we obtain. So we cannot represent ignorance after all, at least not in standard statistical problems. One response to this is that one is never really ignorant. But this is not very convincing. Another is that one cannot get something from nothing. This is better and even sounds more poetic, but it is still a let-down. What a shame that we cannot drink from the grail. Of course, Walley can't be faulted for this—it is just the way sets of probabilities are.

I have barely touched on the rich collection of topics that this book covers. Space does not permit a detailed discussion of each topic, but the chapter titles give a feel for the breadth of topics:

1. Reasoning and Behavior
2. Coherent Previsions
3. Extensions, Envelopes, and Decisions
4. Assessment and Elicitation
5. The Importance of Imprecision
6. Conditional Previsions
7. Coherent Statistical Models
8. Statistical Reasoning
9. Structural Judgments

All this is followed by more than 100 pages of notes and about 50 pages of appendixes. In a book this size, inevitably there are some things that the reader will disagree with, but most ideas have been carefully thought out. Walley often adopts an unusual and provocative stance on issues that are dear to many. For example, he rejects De Finetti's reduction of frequency and epistemic probability in favor of distinct types of probability. Indeed, he chastises Bayesians for making cavalier exchangeability judgments, which he argues require very strong prior information. The reader may not be convinced that the theory carries over so easily to imprecise utilities as well (Seidenfeld, Schervish, and Kadane 1990). One thing missing is a serious treatment of real data analysis examples. There are some examples (including an appendix on eliciting upper and lower probabilities for the 1982 football World Cup), but I prefer books that are peppered with real examples throughout, in the style of Jeffreys and Fisher.

Whether a reader agrees or disagrees with Walley's viewpoint, he or she will learn a lot by reading *Statistical Reasoning With Imprecise Probabilities*. Even when I disagreed on a point, I found I gained much by having my assumptions challenged. The book is full of interesting material on many topics besides upper and lower probabilities. And for recreant Bayesian souls who worry about their priors, the book is essential reading.

LARRY WASSERMAN
Carnegie Mellon University

REFERENCES

Berger, J. O. (1984), "The Robust Bayesian Viewpoint," in *Robustness of Bayesian Analyses*, ed. J. B. Kadane, Amsterdam: Elsevier Science.

——— (1990), "Robust Bayesian Analysis: Sensitivity to the Prior," *Journal of Statistical Planning and Inference,* 25, 303–328.
DeRobertis, L., and Hartigan, J. A. (1981), "Bayesian Inference Using Intervals of Measures," *The Annals of Statistics,* 9, 235–244.
Fine, T. (1988), "Lower Probability Models for Uncertainty and Nondeterministic Processes," *Journal of Statistical Planning and Inference,* 20, 389–411.
Good, I. J. (1962), "Subjective Probability as the Measure of a Nonmeasurable Set," in *Logic, Methodology and Philosophy of Science,* eds. E. Nagel, P. Suppes, and A. Tarski, Stanford, CA: Stanford University Press, pp. 319–329.
Huber, P. J. (1973), "The Use of Choquet Capacities in Statistics," *Bulletin of the Institute for International Statistics,* 45, 181–191.
Huber, P. J., and Strassen, V. (1973), "Minimax Tests and the Neyman–Pearson Lemma for Capacities," *The Annals of Statistics,* 1, 251–263.
Levi, I. (1985), "Imprecision and Indeterminacy in Probability Judgment," *Philosophy of Science,* 52, 390–409.
Pericchi, L. R., and Walley, P. (1991), "Robust Bayesian Credible Intervals and Prior Ignorance," *International Statistical Review,* 59, 1–24.
Schervish, M. J., Seidenfeld, T., and Kadane, J. (1984), "The Extent of Nonconglomerability of Finitely Additive Probabilities," *Z. Wahr. v. Gebiete,* 66, 205–226.
Seidenfeld, T., Schervish, M., and Kadane, J. (1990), "Decisions Without Ordering," in *Acting and Reflecting,* ed. W. Sieg, Dordrecht: Kluwer Academic, pp. 143–170.
Seidenfeld, T., and Wasserman, L. (1993), "Dilation for Sets of Probabilities," submitted to *The Annals of Statistics.*
Wasserman, L., and Kadane, J. (1992), "Symmetric Upper Probabilities," *The Annals of Statistics,* 20, 1720–1736.

Introduction to Stochastic Processes and Their Applications.

Petar Todorovic. New York: Springer-Verlag, 1992. xiii + 289 pp. $49.95.

Let me say it at the beginning: I like this book. It is an introduction to stochastic processes for graduate students with a background in measure and integrability theory and with an intermediate knowledge of probability theory. The book is written in a very clear style that readily directs the student's attention to the main points of interest. The text is restricted to the essentials; no word or formula seems superfluous. The proofs are transparent, leaving unimportant details to the reader. The examples are chosen to be simple and to emphasize the essential properties of stochastic processes. Numbering and notation are reasonable and consistent and do not force the reader to search for the definition of an epsilon, say, 200 pages before. One can always be sure that the basic process is $\xi(\cdot)$ and the covariance function is $C(\cdot, \cdot)$, for instance.

These features make the book attractive for classroom purposes. It does not aim to be a dictionary for the researcher, and it does not intend to cover a whole special topic of stochastic processes.

The book provides the necessary instruments to enable graduate students to understand the general nature of stochastic processes and to attack more special problems in research or in practice.

Chapter 1, "Basic Concepts and Definitions," introduces fundamental notion like equivalent processes, separability, measurability and continuity. It is a short chapter (30 pp.) that leaves out tedious calculations (e.g., the proof of Kolmogorov's consistency theory), but nonetheless it introduces the reader to Brownian motion, the Markov property, and stationarity and makes it all plausible with examples from engineering, physics, and queueing theory.

Although it is possible to jump from Chapter 1 to any other, it is recommended to study Chapter 5 ("L_2 Spaces") before Chapters 6, 7, and 9. Chapter 2, "The Poisson Process and Its Ramifications," is a short introduction to point processes and their applications.

Chapter 3, "Elements of Brownian Motion," and Chapter 4, "Gaussian Processes," direct the reader's attention to processes based on L_2 theory. Chapter 3 presents continuity and sample paths properties as well as various functionals of Brownian motion, concentrated around the distribution of hitting times of Brownian motion. Chapter 3 introduces the Ornstein–Uhlenbeck process, motivated by Langevin's equation, and briefly considers stochastic integrals with respect to Brownian motion. Chapter 4 presents a more general view of stochastic integration and introduces differentiation in quadratic mean. The chapter's main features are stationary and Markov Gaussian processes.

Chapter 5 gives a very readable short introduction to Hilbert space theory and linear operators on L_2. It is supplemented by Chapter 6, "Second Order Processes," which clarifies the interrelationship between properties of the covariance function and behavior of the stochastic process.

Chapter 7, "Spectral Analysis of Stationary Processes," provides an outline of the classical spectral theory, including such important problems as the representation of a stationary process as a stochastic integral, prediction, filtering, Wold decomposition, and rational spectral densities.

Chapter 8, "Markov Processes I," and Chapter 9, "Markov Processes II: Application of Semigroup Theory," give an introduction to the field of Markov processes on 60 pages. Properties of special Markov processes were dealt with in earlier subsequent chapters. In Chapter 8 the classical theory of Markov chains is handled and the notions of strong Markov property and homogenous diffusion are introduced. Chapter 9 continues with the more sophisticated semigroup approach.

Chapter 10, "Discrete Parameter Martingales," gives a brief introduction to martingales. The sections are concentrated mainly around the martingale convergence theorem.

The book contains some minor errors concerning labelling and referencing, but these do not affect readability.

I recommend *Introduction to Stochastic Processes and Their Applications* as an excellent standard text for graduate students in statistics or probability theory. The interested student will benefit from this well-constructed introduction to a nontrivial topic like stochastic processes. Numerous exercises and examples illuminate the contents and support the student's learning process. In addition, references direct the interested reader to the basic literature of the more specialized topics corresponding to each chapter.

THOMAS MIKOSCH
Victoria University of Wellington

Advances in Probability Distributions With Given Marginals.

G. Dall'Aglio, S. Kotz, and G. Salinetti (eds.). Boston: Kluwer Academic Publishers, 1991. xvii + 231 pp. $94.

Fitting univariate distributions to data and to physical phenomena is an important part of data analysis. But the extension to bivariate distributions has been explored only in some special cases. Indeed, because there is no unique extension of a univariate distribution to a bivariate version in which the marginal distributions are given, there have been many bivariate versions, each with its own special characteristics.

A foundation stone of bivariate distributions with given marginals is a set of bounds due jointly to W. Hoeffding and M. Fréchet, which asserts that every bivariate cumulative distribution function lies between two bounds (each a bivariate distribution) determined only by the marginal distributions. This result can be proven in various ways, and there are a number of proofs in the literature. A tantalizing problem is how to extend this bivariate result to higher dimensions or to the imposition of constraints, such as a given correlation, unimodal, and so forth.

The "Symposium on Distributions with Given Marginals" was organized by the Department of Statistics, University LA SAPIENZA, Rome, and was dedicated to the memory of Giuseppe Pompily. There were close to 100 participants and more than 35 paper presentations; of these, 11 were selected as the contents of this volume.

The papers are presented as follows: G. Dall'Aglio begins by providing the historical setting for this topic. In particular, the history of some of the Italian researchers places the early work in perspective. B. Schweizer complements the history of this subject, clarifying the introduction of copulas (uniform distribution with uniform marginals) by A. Sklar. Because the origins here are different from probabilistic origins, this paper is important for understanding the mathematics of copulas. R. B. Nelson clarifies relations between copulas and measures of association (e.g., correlations and other measures of dependency). M. J. Frank investigates the structure of convolutions in terms of copulas, and P. Miksiński, H. Sherwood, and M. D. Taylor discuss probabilistic interpretations of copulas.

S. Kotz and J. P. Seeger review the concept of a density weighting function due to Fréchet and show how some well-known families of distributions can be obtained from this concept. M. H. Metry and A. R. Sampson provide a survey of partial orderings for positive dependence, which points out the relation of some well-known orderings.

H. G. Kellerer investigates the existence of a finite Borel measure with given marginals. Even in the simplest case, the existence of a trivariate distribution $F(x, y, z)$ given two-dimensional marginals $F(x, y, \infty)$ and $F(\infty, y, z)$ does not lead to a simple result. L. Rüschendorf gives a detailed and extensive review of Fréchet bounds, relating the problem to the transportation problem of linear programming. V. Benes and J. Stepan give a further survey on the extreme points of the set of solutions and relate this problem to the extreme points of doubly stochastic matrices.

Finally, S. Cambanis investigates the Eyraud–Farlie–Gumbel–Morgenstern family in the context of random processes and shows that there are problems in the extension to stationary processes.

In summary, the many review papers in this volume provide a survey of the current status of this area of research. *Advances in Probability Distri-*